

ROSEVILLE COLLEGE

YEAR 12

EXTENSION 1 MATHEMATICS

JUNE 2005 ASSESSMENT

Time allowed: 45 minutes+ 2 minutes reading time

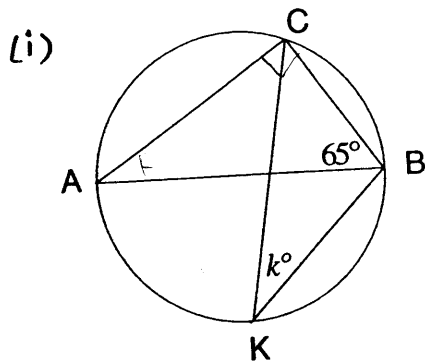
DIRECTIONS TO CANDIDATES:

- **Attempt all questions**
- **All questions are of equal value. The part marks for each section are shown on the right hand side of the page**
- **Please start each question on a new page.**
- **Staple the questions separately**
- **All necessary working should be shown. You may not be awarded the marks for an answer unsupported by working.**

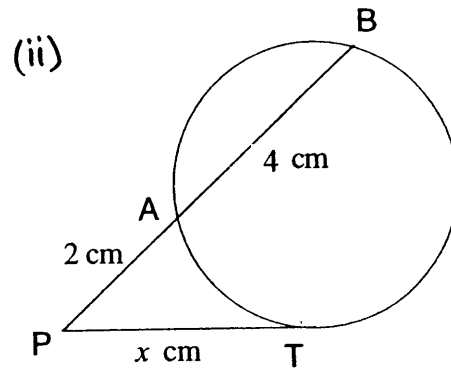
Question 1 (12 marks)

Marks

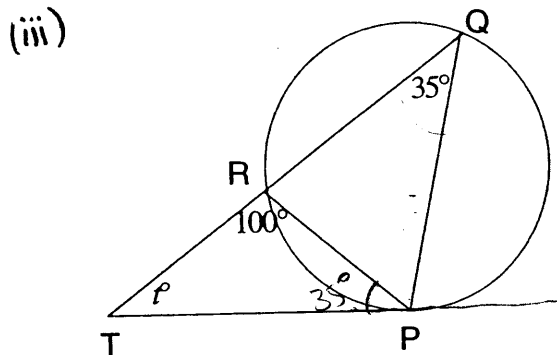
a) Find the value of the pronumeral in each of the following. You do not have to give reasons.



AB is a diameter: $\angle ABC = 65^\circ$,
 $\angle BKC = k^\circ$
 Find the value of k



PT is the tangent at T; $PA = 2$ cm,
 $PT = x$ cm and $AB = 4$ cm. Calculate
 in simplest form, the exact value of x



TP is the tangent at P;
 $\angle PRT = 100^\circ$; $\angle PQT = 35^\circ$
 Find the value of t .

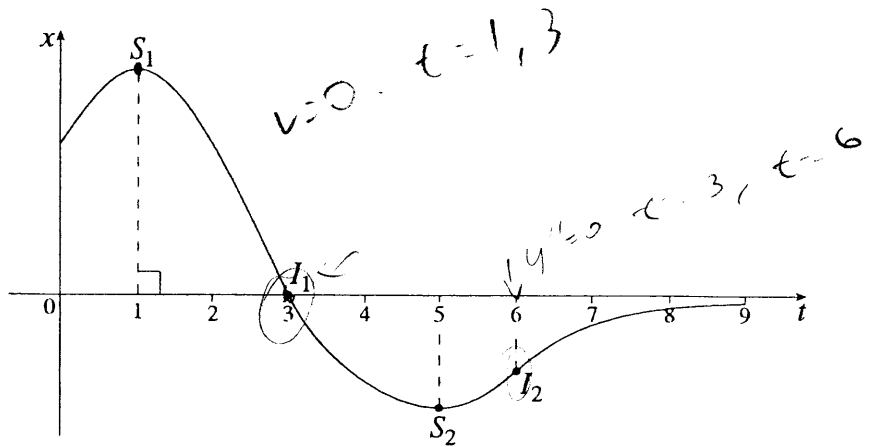
(3)

24 → b) In a fish hatchery, the fish population, N , satisfies the equation $N = N_0 e^{kt}$ where N_0 and k are constants and t is measured in months.

- (i) Initially there were 1 000 fish in the hatchery and at the end of 5 months there were 5000. Find the value of k correct to 3 decimal places. (2)
- (ii) Find the number of fish in the hatchery at the end of 8 months. (Give your answer correct to the nearest hundred.) (1)
- (iii) At the end of which month will the fish population exceed 50 000 for the first time? (1)
- (iv) At what rate is the population increasing at the end of six months? (Give your answer correct to the nearest hundred fish per month.) (2)

6 month
 $t = 6$

2M



The graph shows the position of a particle, moving on a straight line, over the first nine seconds of the motion.

S_1 and S_2 are stationary points; I_1 and I_2 are points of inflexion.

(i) State the times, or periods of time, for which

1. the particle is stationary
2. the velocity is negative
3. the acceleration is positive.

$a > 0$

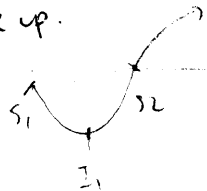
(1)

(1)

(1)

$\downarrow$
 $y'' > 0$

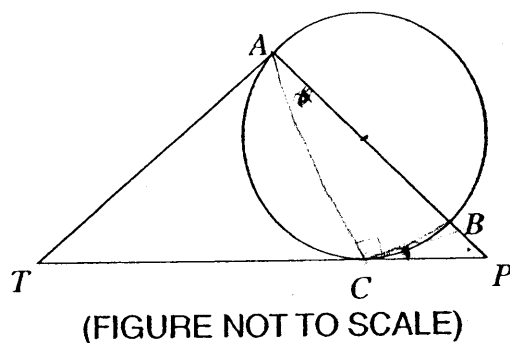
where concave up.



Question2 (12 marks)

Marks

a)



AB is a diameter of a circle ABC . The tangents at A and C meet at T . The lines TC and AB are produced to meet at P . Copy the diagram into your examination booklet. Join AC and CB .

- i) Prove that $\angle CAT = 90^\circ - \angle BCP$. (2)
- ii) Hence, or otherwise, prove that $\angle ATC = 2\angle BCP$. (2)

b) A particle moves along a straight line about a fixed point O so that its velocity, $v \text{ ms}^{-1}$, at time t seconds is given by $v = 4 \sin \left(2t + \frac{\pi}{6} \right)$.

Initially the particle is $\sqrt{3}$ metres to the left of O .

- i. Find expressions for the displacement, x , of the particle at any time t . (2)
- ii. At what time does the particle first return to its initial position? (2)

c) A spherical balloon is being inflated at the rate of $1000 \text{ cm}^3 \text{ s}^{-1}$. You are given that $V = \frac{4}{3}\pi r^3$ and $A = 4\pi r^2$.

- (i) Show that $\frac{250}{\pi r^2}$ is an expression for the instantaneous rate of change of the radius. (2)
- (ii) Find the rate of change of the surface area of the balloon when the radius is 10 cm. (2)

Question 3 (12 marks)**Marks**

- a) A particle moving in a horizontal straight line is performing Simple Harmonic Motion. At time t seconds its displacement x metres from a fixed point O on the line is given by $x = 3 \cos 2t + \sin 2t$, where displacements to the right of O are positive.

Explain whether the particle is initially moving to the right or to the left, and whether it is speeding up or slowing down.

(2)

- b) A particle is moving in simple harmonic motion with a velocity (in m/s) given by $v^2 = 2 - x - x^2$ where x is the displacement (in metres) from a point O .

(i) What are the end points of the particles oscillation?

(1)

(ii) Find the maximum velocity of the particle.

(1)

- c) A particle moves in a straight line so that at time t , ($t \geq 0$) its acceleration a , is given by

$$a = 4x$$

where x is the displacement of the particle from the origin. The particle starts its journey one metre to the right of the origin (at $x = 1$) with a velocity of $v = -2$.

- (i) Show that $v = -2x$.

$t=0, x=1$
 $v=-2$
(2)

- (ii) Express x as a function of t .

(2)

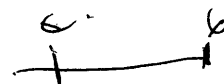
- (iii) Explain whether or not the particle ever moves to the left of the origin.

(1)

- d) A point moving with simple harmonic motions starts from rest at a point 6 cm from the centre of the motion. If the point has a speed of 10 cm/s when it passes through the centre of motion, find the period of the motion.

(3)

$v=0, x=6$



QUESTION 3

c) $x = 3 \cos 2t + \sin 2t$
 $v = -6 \sin 2t + 2 \cos 2t$

$t=0, v = -6 \sin 0 + 2 \cos 0$
 $= 2$

$\therefore$ moving to the right. $\frac{1}{2}$

$\ddot{x} = -12 \cos 2t - 4 \sin 2t$

$t=0, \ddot{x} = -12 \cos 0 - 4 \sin 0$
 $= -12$

v and $\ddot{x}$ have opposite signs

$\therefore$ slowing down. $\frac{1}{2}$

b) i) $v^2 = 2 - x - x^2$

$0 = (2+x)(1-x)$. ($v=0$ at end points.)

$x = -2, 1$

end points are -2 and 1 .

ii) max v when at centre. i.e. $x = -\frac{1}{2}$

$v^2 = 2 - (-\frac{1}{2}) - (-\frac{1}{2})^2$

$= 2 + \frac{1}{2} - \frac{1}{4}$

$= 2\frac{1}{4} = \frac{9}{4}$

$v = \frac{3}{2} \text{ m/s.}$

c) i) $\ddot{x} = 4x$

$\frac{d}{dx}(\frac{1}{2}v^2) = 4x$

$\frac{1}{2}v^2 = 2x^2 + C$

$v = -2, x = 1, \quad \frac{1}{2}(-2)^2 = 2(1)^2 + C$
 $2 = 2 + C$

$C = 0$

$\frac{1}{2}v^2 = 2x^2$

$v^2 = 4x^2$

$v = \pm \sqrt{4x^2}$

$= \pm 2x$ but $v \neq 0$ when $x = 1$

$\therefore v = -2x$

ii)

$$v = -2x$$

$$\frac{dx}{dt} = -2x$$

$$\frac{dt}{dx} = \frac{1}{-2x}$$

$$t = -\frac{1}{2} \ln x + c.$$

$$t=0, x=1, 0 = -\frac{1}{2} \ln 1 + c.$$

$$c=0.$$

$$t = -\frac{1}{2} \ln x$$

$$-2t = \ln x$$

$$e^{-2t} = x.$$

iii)

$$\text{If } t > 0, e^{-2t} > 0$$

$$\therefore \text{No.}$$

d)

$$x = a \cos(nt + \alpha)$$

$$t=0, v=0, x=6$$

$$t=0, x=6, 6 = a \cos \alpha.$$

$$v = -a n \sin(nt + \alpha)$$

$$t=0, v=0 0 = -a n \sin \alpha.$$

$$0 = \sin \alpha.$$

$$\alpha = 0$$

$$\therefore 6 = a \cos 0$$

$$6 = a \times 1$$

$$a = 6$$

$$\therefore x = 6 \cos nt$$

$$v = -6n \sin nt$$

$$x=0, v=10, 0 = 6 \cos nt$$

$$0 = \cos nt \therefore nt = \frac{\pi}{2}, \frac{3\pi}{2} \therefore t = \frac{\pi}{2n}, \dots$$

$$10 = -6n \sin nt$$

$$10 = -6n \sin n \cdot \frac{\pi}{2n}$$

$$10 = -6n \sin \frac{\pi}{2}$$

$$10 = -6n \times 1$$

$$\frac{10}{-6} = n$$

$$n = \frac{5}{3} \therefore \text{Period} = \frac{2\pi}{5/3} = \frac{6\pi}{5}$$